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Electrochimica Acta 51 (2006) 5731–5737
ELECTROCHIMICA Acta
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Electrochemical studies of LiB compound as anode material for lithium-ion battery

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Received 23 August 2005; received in revised form 1 March 2006; accepted 3 March 2006
Available online 18 April 2006

Abstract

Electrochemical studies of LiB compound were carried out for its application as anode for lithium-ion battery. The compound exhibited a reversible discharge–charge behavior between 0 and 0.75 V versus Li/Li^+ with a first discharge capacity of 293 mA h g^{-1}. Discharging to 1.0 V, the first discharge capacity of LiB compound was 660 mA h g^{-1}, but a part of this capacity was irreversible. Impedance spectra were measured at several potentials corresponding to different discharge plateaus. The impedance spectra obtained below and above 0.8 V presented significantly different features. The solid electrolyte interphase layer (SEI) was formed below 0.8 V and assumed a good performance of LiB electrode in this potential range. The SEI was found to deteriorate above 0.8 V, which might be associated with the irreversible discharge capacity.

Keywords: LiB compound; Lithium-ion battery; Anode material; Impedance spectrum; Lithium intercalation

1. Introduction

Various carbon-based systems including graphite, coke and poorly crystalline carbons have been utilized as anode materials for commercial Li-ion batteries. The highest theoretical capacity of graphite is of one lithium per six carbons (LiC6 372 mA h g^{-1}), which results in a volumetric capacity of 800 mA h ml^{-1}. Although these materials fulfill some requirements for use in commercial cells, they are subject to some limitations. These aroused interest in finding alternative negative insertion electrode materials to improve high rate capability and cycle ability [1,2].

Since 1990s, LiB compound has been extensively studied as an anode material for thermal battery [3–5]. The crystal structure and morphology were investigated [6–8]. It was found that the boron atom in the compound are covalently bonded to each other in the <1 1 1> direction forming one-dimensional chains. The morphology of the compound is fibrous. After rolling, the fibers of the composite of metal lithium with LiB compound align along the rolling direction. The LiB belongs to the hexagonal system with B atoms occupying the (2b) site and Li atoms the (2c) site in the unit cell. The LiB alloy shows a physico-chemical property similar to the graphite material which may lead to a similar intercalation/deintercalation behavior for lithium-ion as the carbon materials. LiB shows a good conductivity ($1.43 \times 10^3\ \Omega^{-1}\ cm^{-1}$) and a diffusional rate comparative to that in the metallic lithium.

The theoretical specific capacity of LiB compound from calculation is 452 mA h g^{-1}, much higher than 372 mA h g^{-1} of graphite anode material. In the thermal battery, the discharge curve of LiB alloy electrode presents two plateaus which correspond to the discharge of the free lithium and of the lithium in matrix of LiB compound. It was found that 30% of the lithium in the matrix can contribute to the discharge process. With aim at extending its application, some work has been reported for the charge and discharge behavior of LiB in non-aqueous medium [9,10].

The present work has been done to study the electrochemical behavior of the LiB compound for its potential application as anodic material in lithium-ion battery.

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doi:10.1016/j.electacta.2006.03.023

2. Experimental

2.1. Preparation of LiB

The LiB compound was prepared in an iron crucible with 0.6 g of lithium (purity >99.9%) and 0.4 g of crystalline boron (purity >99 wt.%) corresponding to a composition of near 1:1 for lithium:boron with a slight excess of lithium to avoid lithium-lacking structure. The synthesis, consisting of two exothermic reactions (around 330 and 530 °C) was performed under argon atmosphere. The temperature was carefully controlled to prevent the second reaction ignition by the first exothermic reaction [11].

2.2. Electrochemical measurements

The electrochemical properties of the LiB compound were investigated in the electrolyte of 1 M anhydrous $LiPF_6$ in a 1:1 mixture of ethylene carbonate (EC) and dimethyl carbonate (DMC).

The working electrode was prepared by mixing the LiB powder with 10 wt.% PTFE as binder followed by pressing the mixture onto a nickel mesh to form a tablet (1 cm^2). The discharge–charge behavior was determined in two-electrode cells, using a lithium sheet as counter-electrode; the impedance measurements were carried out in a three-electrode cell with metallic lithium as reference electrode. The potential values in this paper are relative to the Li/Li^+ potential. The electrodes and cells were prepared in grove tank under a dry argon atmosphere.

The discharge–charge tests were controlled with Land model battery testing system. The impedance measurements were carried out by a CHI660A electrochemical station. The amplitude of the alternating current signal was 5 mV over the frequency range between 100 kHz and 1 mHz. All measurements were conducted at room temperature. The impedance data were analyzed with the software of Boukamp [12].

3. Results and discussions

3.1. Discharge/charge behaviors

The samples for the discharge–charge cycling contained 9.60–11.0 mg of active material with a composition of 60 wt.% Li–40 wt.% B. Fig. 1 presents the discharge curve of first cycle of a sample under constant current of 0.2 mA.

A long line near 0 V is observed on the discharge curve corresponding to a capacity of 10 mA h which is attributed to the discharge of free Li in excess in the prepared LiB compound. In addition, the discharge curve presents several plateaus at 0.48, 0.72 and 0.82 V, which may imply that in this LiB compound lithium exists in different kinds of sites with different energies.

Sanchez and Belin reported their experimental study on the charge-discharge behavior of LiB electrode in non-aqueous electrolyte [9]. The LiB compound was found to have some discharge capacity but its structure decayed rapidly after several cycles. In our present work, the LiB electrode exhibited much better performance as shown by the discharge/charge curve of 16 cycles between 0 and 0.8 V depicted in Fig. 2. In this rather low voltage

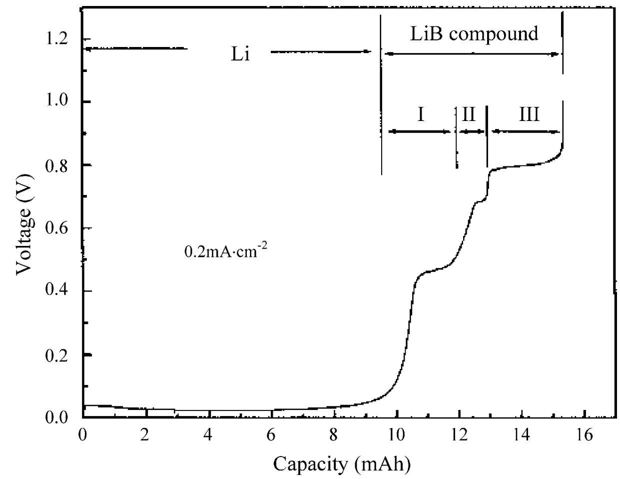

Fig. 1. First discharge curve of LiB/Li battery between 0 and 0.8 V at 0.2 mA.

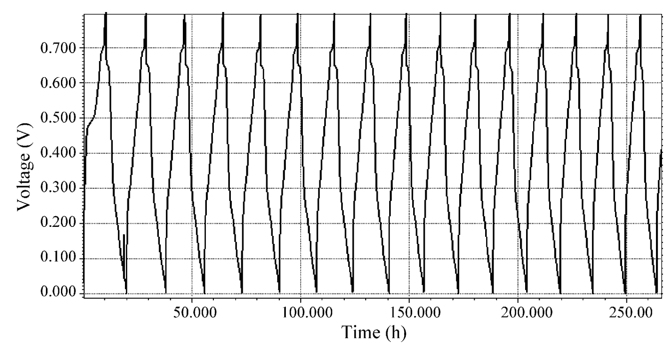

Fig. 2. Discharge–charge curve of LiB/Li battery between 0 and 0.8 V at 0.2 mA.

range, the electrode shows a quite good stability. The evolution of discharge capacity with cycle number is given in Fig. 3.

The discharge capacity in the first cycle was 2.16 mA h corresponding to a specific capacity of 226 mA h g^{-1}LiB. After 15 cycles, the discharge capacity dropped to 1.43 mA h corresponding to a specific capacity of 149 mA h g^{-1} (66.2% of the initial value). The efficiency of charge/discharge was decreased from 98.3 to 94.6%.

The discharge–charge performance of the LiB electrode was examined in an extended voltage range of 0–1.0 V. Fig. 4 shows the curve obtained in the discharge–charge procedure where the

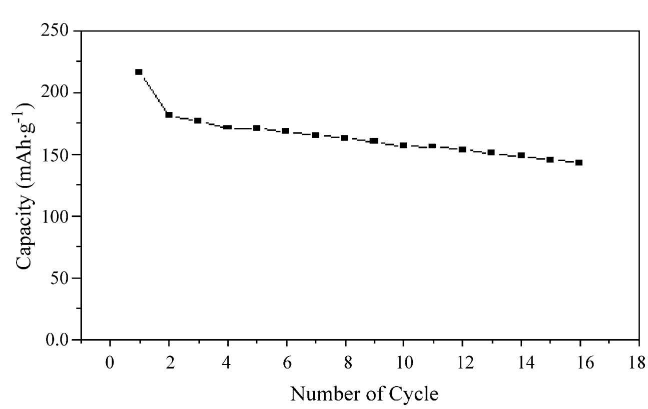

Fig. 3. Cycling performance of LiB/Li battery between 0 and 0.8 V at 0.2 mA.

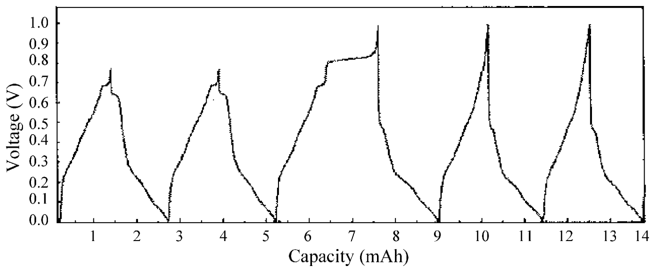

Fig. 4. Discharge–charge curve of LiB/Li battery between 0 and 1.0 V at 0.2 mA.

first two cycles were conducted between 0 and 0.8 V and the following three cycles between 0 and 1.0 V.

The discharge curve of the third cycle gives a long plateau at about 0.85 V which disappears on the following discharge curves (the 4e and 5e cycles). On the discharge curve of the third cycle, we do not see any shoulder related to the discharge plateau at 0.85 V. This means that the lithium deintercalated from LiB structure during discharging at 0.85 V or higher potential cannot return to their original sites, this part of lithium being irreversible. But it can be seen from the curves of the two following cycles that the LiB electrode discharged above 0.85 V maintained part of its reversible discharge/charge capacity.

The discharge/charge behavior of the first cycle was listed in Table 1. It is seen from Table 1 that when the discharging voltage was extended from 0.8 to 1.0 V, the discharge capacity was much increased but a large part of this capacity was irreversible. If the capacity of 660 mA h g^{-1} could be recovered, it would be of great significance for LiB to be applied as anode material for lithium-ion battery.

3.2. Impedance investigation

In order to elucidate the different behaviors of LiB compound at different discharging potentials, especially the mechanism of the irreversibility above 0.8 V, electrochemical impedance spectroscopy (EIS) studies were performed.

EIS measurements were carried out in potential ranges: (1) below 0.44 V (0.28, 0.34 and 0.40 V); (2) 0.44–0.66 V (0.49, 0.66 V); (3) 0.66–0.80 V (0.75 and 0.8 V); (4) above 0.80 V (0.95 V).

3.3. Below 0.44 V (0.28 and 0.44 V)

Fig. 5 depicts the impedance spectra of LiB electrode at 0.28 V. The Nyquist plot shows an inductive arc in high frequencies followed by two capacitive semi-circles in medium frequencies. In low frequency zone, a straight line with a slop

Table 1
Discharge–charge capacities of LiB/Li battery of first cycle

Voltage range	Discharge capacity (mA h g^{-1})	Charge capacity (mA h g^{-1})
0–0.6	220	210
0–0.8	293	274
0–1.0	660	300

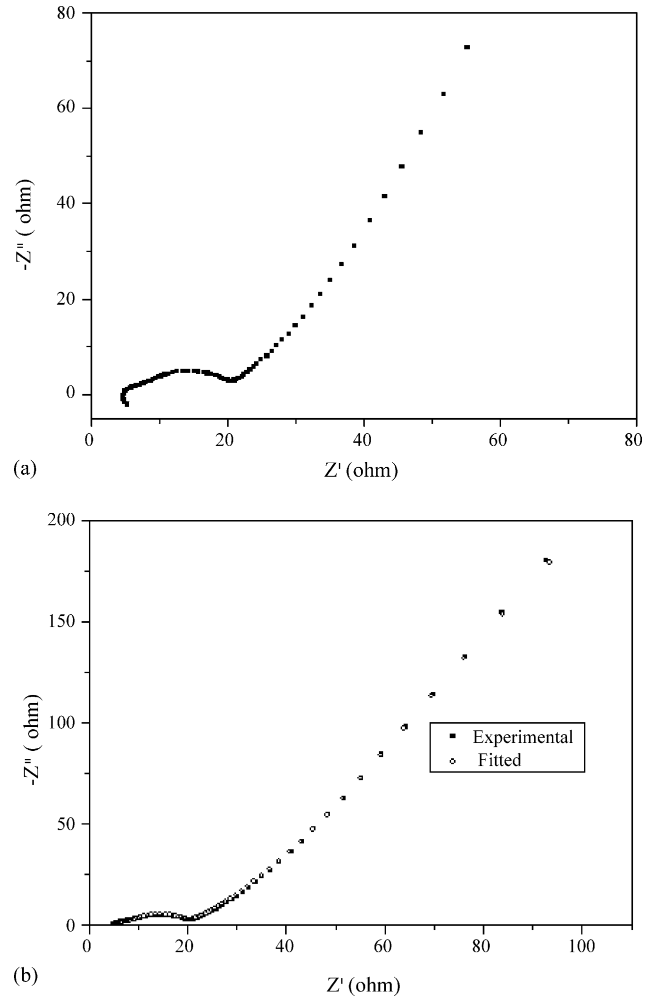

Fig. 5. Nyquist plots of LiB electrode at 0.28 V: (a) experimental plot; (b) fitted plot.

of about 45° is observed. These are the typical features of the electrode in lithium-ion battery [13]. The lithium intercalated graphite anode has been extensively studied by EIS. It is reported that its impedance spectrum exhibited two semicircles with the inclined line. The semicircle in the high frequency region was assigned to the SEI film resistance and that in lower frequency zone to the charge transfer resistance [14]. In a recently reported study of lithium phosphorous oxynitride (LiPON) as a passive layer for anodes in lithium secondary batteris, the impedance spectrum with two semicircles was also observed on the Pt electrode with LiPON layer [15]. In our present work, the first RC semicircle can be attributed to the passive film formed at the anode/electrolyte interface and the semi-circle is less affected by the potential; the second semi-circle in the medium frequencies represents the charge transfer process at the electrode surface with its diameter changing with the electrode potential. The straight line is a feature of Warburg diffusional impedance.

The passive layer at the graphite electrode surface is generally known as solid electrolyte interphase (SEI) which is formed by the decomposition of electrolyte at the electrode/electrolyte interface. The SEI layer protects the electrode from the attack of electrolyte and prevents the organic species entering the struc-

Table 2
Impedance parameters obtained at 0.28–0.75 V, using $LR_{\Omega}(C_fR_f)(C_{dl}(OR_{ct}W))$

Potential (V)	L ($\times 10^{-6}$ H cm^2)	R_{Ω} (Ω cm^2)	C_f ($\times 10^{-6}$ F cm^{-2})	R_f (Ω cm^2)	C_{dl} ($\times 10^{-6}$ F cm^{-2})	O[a]		R_{ct} (Ω cm^2)	Warburg, Y_O (S s$^{0.5}$ cm^{-2})
						Y_O (S s$^{0.5}$ cm^{-2})	B (s$^{0.5}$)		
0.28	3.48	4.79	11400	1130	6.32	0.0045	0.049	2.72	0.0414
0.34	3.49	4.82	11200	881	6.22	0.004	0.051	2.76	0.0403
0.44	3.19	4.44	8.54	7.12	8.31	0.035	1.92×10^{17}	3.11	381
0.49	3.216	4.86	90.3	9.81	8.19	0.500	0.3131	3.62	0.0361
0.66	2.79	4.79	8.48	3.95	84.5	15.9	0.074	11.3	0.0824
0.75	3.29	4.75	9100	1203	6.00	0.004	0.047	2.90	0.0705

[a] T for 0.44–0.66 V.

ture of electrode hence blocking the pathway of Li species. The SEI layer plays a fundamental role in the stability and cyclability of the graphite electrode [16,17]. The formation of the SEI layer at LiB electrode/electrolyte has been revealed previously by our SEM investigation [11] and is confirmed by the impedance measurement. It is of same importance for the electrochemical performance of LiB electrode as for graphite electrode.

The inductive arc appearing on Nyquist plot in high frequencies is noticeable. The inductance behavior of the lithium-ion battery was reported and attributed to the porous features of the electrodes [13,18,19]. Hence, the value of the inductance L should remain constant during cycling unless the electrodes were mechanically damaged. Some authors even considered that the parallel resistance RL of the inductance did not represent any practical physical process and was only a schematic element for the convenience of the fitting procedure. But in our present work, the inductive feature on the impedance spectra was potential dependent (it was observed in 0–0.75 V but not at potentials above 0.75 V), hence it should be relative to the anodic reaction process. The inductance may be attributed to an adsorption of reactive species on the passive film.

The impedance spectrum obtained at 0.44 V (Fig. 6) shows a behavior similar to that at 0.28 V, implying that below 0.44 V the reaction mechanism, the passive film and the adsorption behavior have no significant differences at this two potentials.

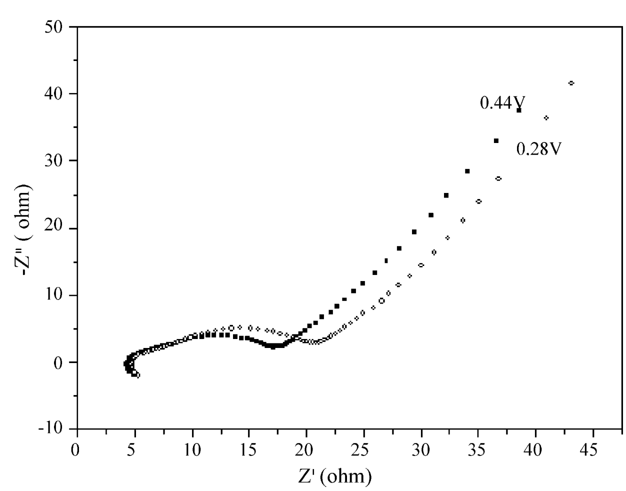

Fig. 6. Nyquist plot of LiB electrode obtained at 0.44 V compared with that at 0.28 V.

3.4. Equivalent circuit fitting

Several equivalent circuits were proposed to fit the experimental impedance data and the best fit was achieved using the circuit $LR_{\Omega}(C_fR_f)(C_{dl}(OR_{ct}W))$, where L is an inductive element; R_{Ω} the ohmic resistances between the working and reference electrodes; the parallel circuit (C_fR_f) represents the diffusional process of species in the passive layer at electrode/electrolyte interface; C_{dl} the double layer capacitance at the electrode surface; W, Warburg and O, Cothyperbol, corresponding respectively to the semi-infinite length diffusion and finite diffusion length diffusion related impedances; R_{ct} is the charge transfer resistance of the Faradaic reaction which is in serial with the two diffusional elements to represent the resistance the lithium ions encounter in the charge transfer and transport processes.

As shown in Fig. 5b, a good fit was achieved for the impedance spectrum at 0.28 V. The parameters obtained are listed in Table 2.

3.5. 0.44–0.66 V zone (0.49 and 0.66 V)

The impedance spectra obtained at 0.44 and 0.66 V are compared in Fig. 7 where an enlarged semi-circle at 0.66 V is seen but the spectra show similar features. Generally, the charge transfer resistance of a reaction decreases with the overpotential. The fact that the RC semi-circle enlarged as the potential changed from 0.44 to 0.66 V implies that the lithium species discharging at this two different potentials were from different sites.

For the impedance spectra at 0.49 and 0.66 V, a better fit was achieved by using the equivalent circuit $LR_{\Omega}(C_fR_f)(C_{dl}(TR_{ct}W))$ where the component cothyperbol O was replaced with Tanthyperbol T. The impedance parameters obtained is given in Table 2. O and T are two diffusion related elements dealing with finite length diffusion. O deals with the case where one boundary imposes a fixed concentration for the diffusing species; T describes diffusion through a medium where one boundary is blocking for the diffusing species, such as a thin mixed conducting electrode.

3.6. 0.66–0.80 V zone (0.75 and 0.8 V)

Fig. 8 shows the impedance spectrum at 0.75 V which exhibits in the medium frequencies a slightly decreased semi-circle cor-

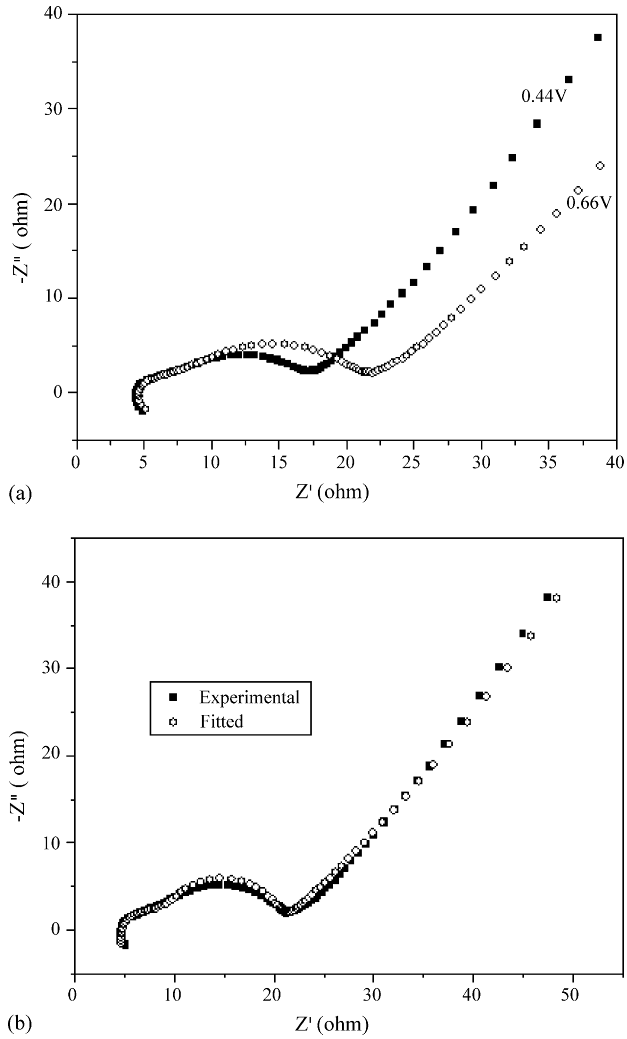

Fig. 7. Nyquist plot of LiB electrode obtained at 0.66 V compared with that at 0.44 V: (a) experimental plot; (b) fitted plot for 0.66 V.

responding to the Faradaic reaction with no significant difference in comparison to that at 0.66 V. These mean that at 0.75 V the electrochemical behaviors (electrode reaction and the SEI layer) of the LiB electrode had not significant changed.

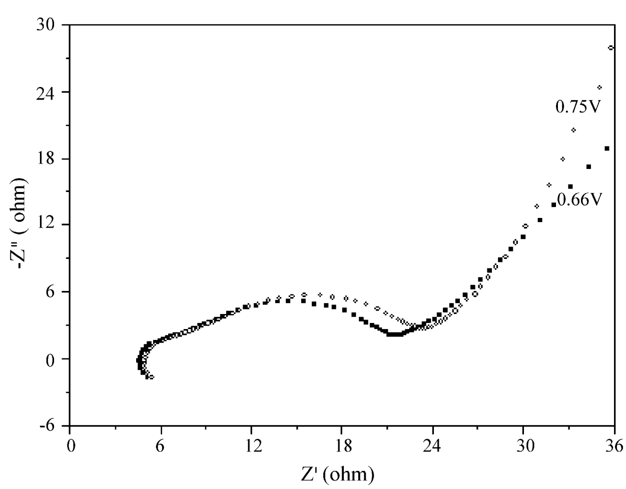

Fig. 8. Nyquist plot of LiB electrode at 0.75 V compared with that at 0.66 V.

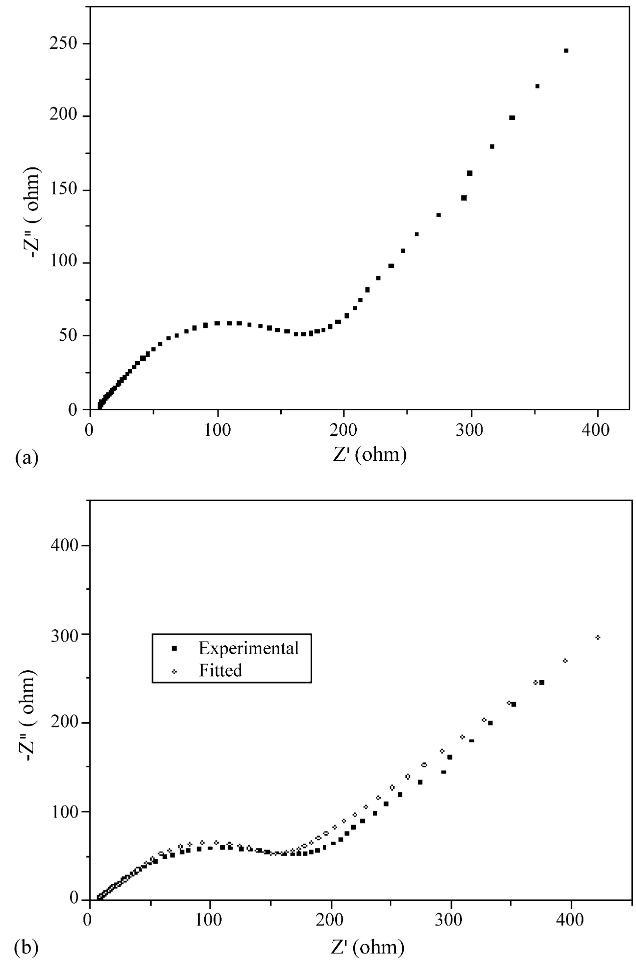

Fig. 9. Nyquist plot of LiB electrode at 0.80 V: (a) experimental plot; (b) fitted plot.

3.7. Above 0.80 V (0.80 and 0.95 V)

As seen from the discharge–charge measurement, the LiB electrode exhibited an irreversibility when discharged at potential higher than 0.8 V. To elucidate the mechanism, impedance spectra were determined at 0.8 and 0.95 V. As shown by the Nyquist plots in Fig. 9, the impedance features are remarkably different from these at lower potentials. The inductive loop in high frequencies disappears, turning to a straight line. In medium frequency range, a single semi-circle is observed instead of two semi-circles, indicating that the semi-circle corresponding to the SEI layer disappeared.

It can be observed that the size of the semi-circles at 0.8 and 0.95 V is much greater than that at 0.75 V. From these behaviors, it can be deduced that (1) the SEI layer at electrode/electrolyte decays at high potentials; (2) the mechanism of the discharge reaction at potentials above 0.80 V differs from that in lower potentials which involves no longer an adsorption process of some kinds of species in the electrode interface.

For the impedance spectra at 0.8 and 0.95 V, a good fit was achieved using a simplified circuit $R_{\Omega}(C_{dl}(TR_{ct}W))$ where the circuit (C_fR_f) representing the passive layer was omitted. The parameters obtained are listed in Table 3.

Table 3
Impedance parameters at 0.80 and 0.95 V fitted using $LR_{\Omega}(C_fR_f)(C_{dl}(OR_{ct}W))$

Potential (V)	R_{Ω} (Ω cm^2)	C_{dl} ($\times10^{-6}$ F cm^{-2})	O		R_{ct} (Ω cm^2)	Warburg, Y_O (S s$^{0.5}$ cm^{-2})
			Y_O ($\times10^{-4}$ S s$^{0.5}$ cm^{-2})	B (s$^{0.5}$)		
0.80	7.64	1.74	7.10	0.076	6.65	0.0023
0.95	12.9	0.82	6.64	0.047	4.32	0.0034

3.8. Impedance features associated with the SEI layer

In potential between 0 and 0.75 V, the spectra are similar: an inductive loop appears in high frequencies followed by two consecutive semi-circles in high and medium frequency range. These features can be interpreted by the stable existence of the SEI layer and an adsorption of reactive or product species in the electrode surface.

The impedance parameters C_f and R_f are related to the SEI layer. As can be seen from Table 2, the values of C_f and R_f vary with electrode potential in 0–0.8 V range: the capacitance C_f is greater at low potential (0.28 and 0.34 V) and it decreases as the potential increases, meaning that the surface area of the SEI layer is reduced at higher potential. At potential above 0.8 V, LiB electrode exhibits significantly different features: the inductive loop attributed to an adsorption and the semi-circle representing the SEI disappears. It should cite here that the value of C_f value is very sensible to the equivalent circuit used in fitting and the difference of C_f values at different potentials is too large to be attributed only to the evolution of SEI layer, a quantitative analysis of SEI based on the C_f value is difficult.

It can be deduced from these results that the SEI layer forms at lower potentials and it reduces as the potential increases; at potential above 0.80 V, the SEI layer decays. It can also be assumed that the discharge process of Li involves different mechanisms at electrode with and without the SEI layer.

The reduction of the SEI layer at higher potential revealed from impedance spectra has been confirmed by our SEM photographs [11]. Some researchers reported that the SEI film on a graphite electrode is formed only when it is polarized from open potentials, and substantial Li intercalation occurs at potentials below the SEI formation potential [15]. Our results may be in some extend in consistence with these in literature but are not enough to explain the evolution of the passive layer with the discharge depth, and to propose a solution to inhibit the decay of the SEI layer at high potential.

3.9. Impedance features associated with the transport behaviors

It can be seen from the values of R_{Ω} in Table 2 that over potentials from 0.28 to 0.75 V, the ionic and mixed conductance in of LiB electrode are not affected by the potential. The ohmic resistance of LiB electrode remarkably increases as the potential increases to 0.8 V (see Table 3).

It has been observed from the Nyquist plots in Fig. 10 that the RC semi-circle much enlarges at 0.80 and 0.95 V implying that the discharge process of Li becomes more difficult with the discharge deep at higher potential. One can notice from Table 3 that the value of the charge transfer resistance R_{ct} does not increases significantly with the potential. It is rather the diffusional performance that worsens at higher potential as indicated by the parameters Y_O for O component.

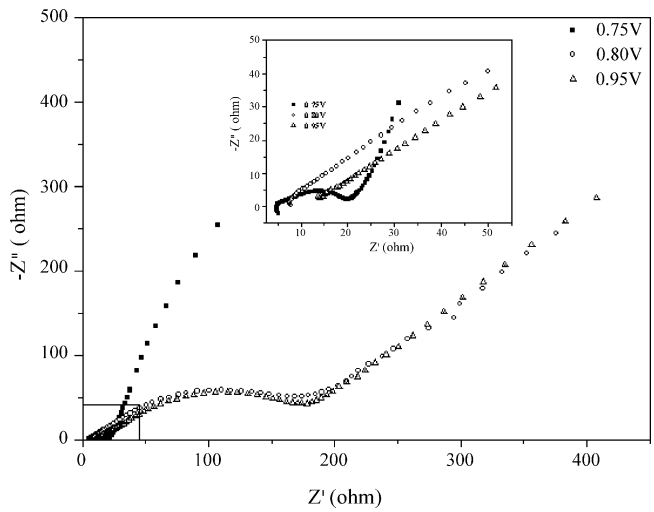

Fig. 10. Nyquist plots of LiB at 0.75, 0.8, and 0.95 V.

The changes in conductance and diffusional performances of LiB anode at higher potentials may be due to a blocking of the pathway for the Li species in the structure which may results from the decay of SEI layer. As part of the discharged Li is not rechargeable, it can not be excluded that a part of matrix collapsed during the deep extraction of Li from its lattices. But it worthies to notice that the charge transfer of Li proceeds normally at 0.8–0.95 V and thereafter the LiB electrode maintains part of its reversible discharge/charge capacity (Fig. 4). This means that the main LiB matrix remains uncollapsed and the SEI layer can be self-repaired at lower potentials.

As revealed by a lot of research works, the formation and stability of SEI layer at electrode/electrolyte is critical for the performance of anode in lithium batteries. The decay of the SEI layer may be responsible for the irreversibility of part of the discharge capacity. The revelation of the mechanism for the decay of the SEI layer and a solution to inhibit this decay will be an important subject of our further work.

4. Conclusions

LiB compound was synthesized and tested as anode material for lithium-ion battery that exhibited an reversible discharge/charge behavior between 0 and 0.75 V with a first discharge capacity of 293 mA h g^{-1}. The discharge/charge capacity decreased progressively with the cycle number. Discharging to 1.0 V, the first discharge capacity of LiB compound was 660 mA h g^{-1}, but a part of this capacity was irreversible.

Impedance measurements were carried out on LiB electrode over potentials from 0.28 to 0.95 V. The spectra obtained at potentials below and above 0.80 V showed remarkable different features. It is revealed that the SEI layer was formed between 0 and 0.75 V and it assured a good performance of the electrode. The SEI layer decayed at potential above 0.80 V which resulted in a blocking of the pathway for the active species in the LiB structure and might in turn be responsible for the loss of reversible discharge capacity.

LiB compound appears to be a potential anode material for lithium-ion battery. However, an improvement of the irreversibility of the discharge capacity is critically needed. An elucidation of the mechanism of how evolutes the passive layer with the electrode potential especially how it decays at higher potential is very interesting subject.

Acknowledgement

The authors thank the National Natural Science Foundation Committee of China for the financial support of this research work (no. 50172062).

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